ECE321 ETHICS CASE STUDY #1

"To Ship or Not to Ship"

The Scenario

The Setting: A mid-sized electronics manufacturer is about to launch a major consumer product containing a custom microcontroller.

The Discovery: During final quality assurance (QA) testing, a junior-level engineer discovers a rare edge-case glitch. Under specific, high-stress operating conditions, the component may miscalculate or overheat, failing roughly 1% to 2% of the time.

The Pressure:
Fixing the hardware bug requires a complete re-spin of the circuit board layout, which will delay the product release by four months and cost the company millions of dollars, potentially missing the critical holiday/market window.

Marketing and upper management argue that 98% of users will never encounter the edge case. They propose shipping the product now and issuing a software patch or quiet RMA (Return Merchandise Authorization) replacement later only for customers who actually complain.

The Engineer's Dilemma: Do you sign off on the shipment, blow the whistle internally (or externally), or refuse to stamp/approve the test results?

Core Ethical Questions for Consideration

- **Utilitarian vs. Deontological Ethics:** Does the financial survival of the company (and jobs for 500 employees) outweigh a small statistical risk of hardware failure? Or does an engineer have an absolute duty to never knowingly ship a flawed safety- or performance-critical product?

- **The Ethics of "Quiet Fixes":** Is it honest to let consumers buy a product with a known hardware defect while hoping they never notice? Where is the line between an acceptable manufacturing tolerance and a hidden defect?

- **Hierarchy and Whistleblowing:** What are the realistic career consequences (e.g., getting blackballed, sidelined, or fired) for the QA engineer standing her ground, and how should professional codes of ethics, like the IEEE code attached, guide their actions?

- **The Intel Pentium Flaw (1994):** How did Intel navigate a similar situation?

IEEE Code of Ethics

We, the members of the IEEE, in recognition of the importance of our technologies in affecting the quality of life throughout the world, and in accepting a personal obligation to our profession, its members and the communities we serve, do hereby commit ourselves to the highest ethical and professional conduct and agree:

I. To uphold the highest standards of integrity, responsible behavior, and ethical conduct in professional activities.

1. to hold paramount the safety, health, and welfare of the public, to strive to comply with ethical design and sustainable development practices, to protect the privacy of others, and to disclose promptly factors that might endanger the public or the environment;

2. to improve the understanding by individuals and society of the capabilities and societal implications of conventional and emerging technologies, including intelligent systems;

3. to avoid real or perceived conflicts of interest whenever possible, and to disclose them to affected parties when they do exist;

4. to avoid unlawful conduct in professional activities, and to reject bribery in all its forms;

5. to seek, accept, and offer honest criticism of technical work, to acknowledge and correct errors, to be honest and realistic in stating claims or estimates based on available data, and to credit properly the contributions of others;

6. to maintain and improve our technical competence and to undertake technological tasks for others only if qualified by training or experience, or after full disclosure of pertinent limitations;

II. To treat all persons fairly and with respect, to not engage in harassment or discrimination, and to avoid injuring others.

7. to treat all persons fairly and with respect, and to not engage in discrimination based on characteristics such as race, religion, gender, disability, age, national origin, sexual orientation, gender identity, or gender expression;

8. to not engage in harassment of any kind, including sexual harassment or bullying behavior;

9. to avoid injuring others, their property, reputation, or employment by false or malicious actions, rumors or any other verbal or physical abuses;

III. To strive to ensure this code is upheld by colleagues and co-workers.

10. to support colleagues and co-workers in following this code of ethics, to strive to ensure the code is upheld, and to not retaliate against individuals reporting a violation.

Adopted by the IEEE Board of Directors and incorporating revisions through June 2020.

Assignment:

After reading the IEEE Code of Ethics and considering the thought exercises listed on page 1, create an analysis of this ethical case. *AI should not be used to either create ideas or to perform the analysis*. The expectation for the analysis is to be less than 2 pages. It should apply no fewer than 3 items from the code. This is an individual assignment but some basic collaboration is allowed. This means you can discuss one line of thinking, say related to safety, health, and welfare. You should consider not just the ethical obligations of the junior engineer, but of senior management as well. Enjoy!